

Disease Defence and Human Health

Today in Canada you can expect to live longer than your great-grandparents. This was not always the case. In the past, many people died from widespread outbreaks of disease. What causes disease? How does disease spread? How does the body protect itself from disease? How do the characteristics you inherit from your parents affect your health? How does the environment influence health?

For centuries, scientists have searched for answers to all of these questions. They discovered that disease-causing agents spread disease. In this unit you will learn about disease-carrying agents such as *Giardia*, which is present in water.

Today, antibiotics treat many diseases successfully. Research about how characteristics are passed from parent to child helps us better understand inherited diseases.

In this unit you will learn about all of these things, and about how public health helps to keep us all healthier than our great-grandparents. Knowledge about these topics can benefit your life in many ways.



9 Disease

Getting Ready...

- What causes disease?
- What can you do to avoid getting a cold?
- How might your lifestyle increase your risk for disease?
- How can you prevent food poisoning?



Your head is throbbing. Your throat is so raw that it hurts to talk. You start to cough.

Before you cough again, consider your actions. The germs that cause colds are so small that you need a microscope to see them. But every time you cough without covering your mouth, you are sharing germs with others nearby.

Disease-causing agents such as cold viruses are all around you. In this chapter, you will investigate the conditions that allow these agents to grow and spread. You will learn about other causes of disease and about the differences between

diseases spread by germs and those that are not caused or spread in this way. You will also find out how lifestyle, diet, and environmental factors such as lack of cleanliness play a role in disease. As you investigate ways to prevent the contamination of food, you will also learn about techniques used to destroy bacteria that contaminate food.

As you understand more about disease, you will see *why* you need to protect yourself from disease — and how to do it. In addition, you will understand how important your personal actions are the next time you start to sneeze!



What You Will Learn

In this chapter you will learn:

- what causes human disease
- how disease spreads
- the difference between communicable and non-communicable diseases
- how to handle food safely
- about techniques used to kill germs

Why It Is Important

- Every year, many people get sick or die from diseases that could have been prevented. Knowing about the causes and spread of disease can help you make better decisions about taking care of your health.

Skills You Will Use

In this chapter you will:

- identify causes of disease
- distinguish between communicable and non-communicable diseases
- investigate techniques used to prevent food contamination
- assess the impact of techniques used to kill germs

9.1 Causes of Human Disease

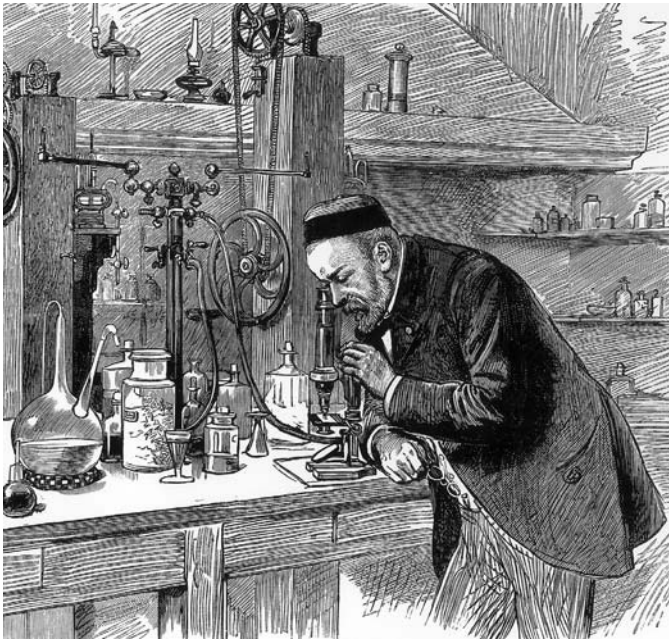


Figure 9.1 In the 1800s, French winemakers were upset because their wine kept turning to vinegar. Louis Pasteur discovered that a microbe was causing the spoilage. He heated the grape juice to kill the microbes. This technique is still used today and is called **pasteurization**.

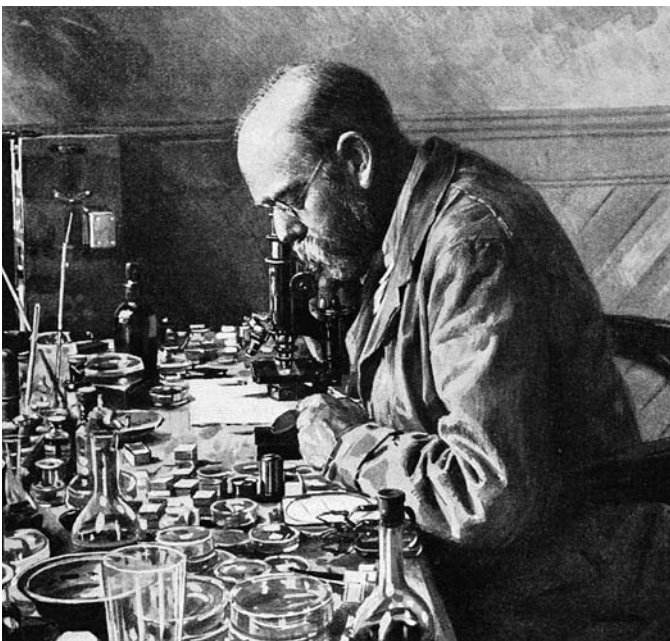


Figure 9.2 Robert Koch identified the pathogen that causes anthrax. He discovered that a pathogen could spread from a sick person to a healthy person.

Why do people get sick? In the past, people thought diseases were caused by various spirits or gods. Some people still hold these ideas. Scientists have another theory.

Disease, or illness, prevents the normal functioning of the body. As a result of the invention of the microscope, scientists like Louis Pasteur and Robert Koch found and identified micro-organisms known as **microbes**. The microbes that cause disease are called **pathogens** or germs.

Pathogens are not the only things that cause disease. Some diseases are caused by a poor environment. For example, air pollution can damage the lungs.

Other diseases are genetic; for example, muscular dystrophy. This disease causes progressive weakness of the muscles.

Disease may also arise from a combination of factors that include the environment, diet, lifestyle, genetics, pathogens, and stress. For example, have you ever noticed how often you get sick when you are worried about a relationship? Often, when people are stressed, they eat poorly and may stop exercising. The diet and lifestyle choices they make combine with stress to make them more likely to get sick.

Pathogens live in toilets, air conditioners, and dirty wash cloths. If these are not cleaned regularly, the number of pathogens may increase. Indoor and outdoor pollutants can affect your health to a greater or lesser degree, depending on the type and amount of pollutants and your sensitivity to them.

READING Check

Identify five factors that may interact with one another and possibly make you sick.

Pathogens

Pathogens are disease-causing microbes. Most pathogens are microscopic. These include some bacteria, viruses, fungi, and protists.

Bacteria

Bacteria are on your skin, in your mouth, in the air you breathe, and in the food you eat. Why can't you see them? Because they are single-celled organisms so small that you need a microscope to see one.

Some bacteria are beneficial to humans. Yogurt, for example, is made from milk and a bacterium that causes milk to go sour.

Other bacteria are harmful to humans. They cause disease and spoil food.

Both the harmful and helpful kinds of bacteria reproduce quickly and form colonies when conditions are good. The box below shows the type of environment bacteria like.

1. **Temperature** — Most bacteria are active at warm temperatures. Refrigeration or freezing slows bacterial growth. Bacteria die when temperatures get too high.
2. **Moisture** — Bacteria need water. Many cannot reproduce without it.
3. **Food** — Bacteria thrive on sugar, protein, or fats.

Viruses

Like bacteria, viruses can be found anywhere there are cells to infect — and you are made of *trillions* of such cells.

Viruses, such as the common cold virus, are much smaller than bacteria. They reproduce by taking over cells in the body. Viruses use the part of the body cell usually used to reproduce cells. In this way, the number of viruses quickly multiplies, and they invade other body cells.

Figure 9.4 The HIV virus that causes AIDS looks like a bomb. It is a virus that flows through the bloodstream of infected individuals. An infected person may contaminate the blood of an uninfected person in one of several ways. These include:

- having sexual intercourse
- sharing intravenous needles
- receiving a transfusion of infected blood
- moving through the birth canal of an infected mother

DidYouKnow?

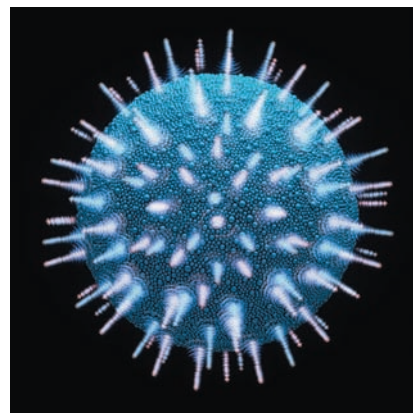
Gingivitis is one of the most common diseases in the world. It is an inflammation of the gums caused by bacterial plaque, a sticky, clear film that forms constantly on your teeth. Brushing and flossing daily is the best way to combat gum disease.



Figure 9.3

READING Check

List the conditions that bacteria need to grow. Explain why each is important.



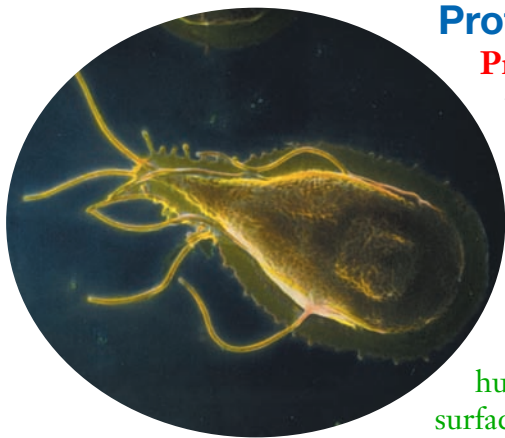


Figure 9.5 *Giardia* is the most common protist in the world. In Canada, it is also called beaver fever because beavers often carry and spread the protist.

Protists

Protists are single-celled organisms that share some characteristics with both animals and plants. Two common protists are the amoeba and the paramecium.

Some protists cause serious diseases such as malaria, African sleeping sickness, and amoebic dysentery. Malaria is spread by mosquitoes, sleeping sickness by Tsetse flies. Amoeba can be picked up from contaminated water or food.

A protist called *Giardia* lives in the intestines of infected humans or animals. *Giardia* may be found in water, food, soil, or surfaces that have been contaminated with the feces from infected humans or animals. You can become infected after you drink the contaminated water.

To prevent getting *Giardia*, drink only safe water.



How do protists spread?

Fungi

Fungi include molds, yeasts, and mushrooms. They cannot make their own food by photosynthesis. Some fungi are **parasites** and can live only on host organisms. Fungi, such as mushrooms taste great on pizza or salad. Others cause disease. Two examples of this are athlete's foot and ringworm, two fungi that live on the body.



Figure 9.6 Athlete's foot is caused by a fungus that lives on the skin. It breeds under warm, moist conditions such as those found in sweaty socks and shoes. To prevent it, wear leather shoes and sandals so your feet can "breathe." Wear sandals or slippers in places such as change rooms with showers.

Key Terms

pasteurization
disease
microbes
pathogens
bacteria
viruses
fungi
protist
parasites

Check Your Understanding

1. Describe three factors that cause disease. Provide an example for each one.
2. What is a pathogen? Give three examples of pathogens.
3. Explain how viruses such as the AIDS virus spread.
4. What might happen to the bacteria in a hamburger you left on the counter for a few hours? Explain why.

9.2 Communicable and Non-communicable Diseases

You can catch a cold from a friend, but can you get diabetes from one?

The answer is no. You can catch a cold from someone because a cold is a communicable disease. Colds are caused by pathogens that can spread from one person to another.

But diabetes is a non-communicable disease. It is not caused by a pathogen. Such diseases cannot spread from person to person.

READING check

List the two general types of disease and explain the difference between the types.

- 1. Communicable diseases** — These diseases are caused by pathogens. The diseases are communicable because the pathogen carrying the disease can spread from one organism to another. The flu, the common cold, and AIDS are all examples of communicable diseases. Table 9.1 shows how such diseases spread.
- 2. Non-communicable diseases** — These diseases are not caused by pathogens. As a result, they cannot be passed from one organism to another.

Interestingly, we do not know the cause of all non-communicable diseases. Today, doctors have a good idea of how people get such non-communicable diseases as diabetes, cancer, and arthritis. They are still not sure what causes depression and multiple sclerosis. But we do know one thing — people who have them do not pass them on to others.

READING check

Describe four ways in which communicable diseases can be transmitted.

Table 9.1 Communicable Diseases and Effects on Health

How Spread	Sample Disease and Type of Pathogen That Causes It	Effects of Disease
Airborne - commonly spread by coughing and sneezing - first sites of infection are the nose, throat, and lungs	(a) common cold — virus (b) influenza (flu) — virus	(a) runny nose, sore throat, cough, low-grade fever (b) fever, aches and pains, cold-like symptoms, extreme fatigue
Water-borne - contaminated water - disease often spreads through feces that contain bacteria. Feces from infected humans or animals can end up in the water system (for example, swimming pools, hot tubs, streams, ponds).	(a) dysentery — bacterium (b) <i>E. coli</i> 0157 — bacterium	(a) severe cramps, diarrhea (often with blood and mucus), fever, vomiting (b) cramps, bloody diarrhea, complications such as kidney failure
Carried in Food - unsafe food handling - improper methods used to preserve food - improper storage of food	(a) botulism (form of food poisoning) — bacterium (b) salmonellosis (form of food poisoning) — bacterium	(a) double vision; difficulty speaking, breathing, and swallowing (b) fever, headache, nausea, vomiting, diarrhea
Direct Contact direct contact with an infected person or with materials handled by an infected person	(a) meningitis — bacterium and virus (b) STDs (sexually transmitted diseases) — bacterium and virus	(a) severe headache, high fever, vomiting, neck stiffness, joint pain (b) vary according to disease but often include sores, irregular discharge, itching

DidYouKnow?

Foods containing chicken may become contaminated with bacteria if they are not handled properly. To prevent **salmonellosis**, chicken needs to be kept hot or cooled rapidly after it is cooked thoroughly. Chicken not eaten right away should be refrigerated.

READING check

How does a clean environment protect you from disease?

The Importance of a Clean Environment

Diseases can be spread by air, water, food, and contact. Why aren't we all sick most of the time?

The answer has to do with our environment. Clean environments reduce the number of viruses and bacteria present. As a result, there are fewer pathogens to make us sick.

Some environments have to be super clean. **There are so many pathogens associated with human feces that bathrooms have to be sanitary. This means maintaining clean, germ-free conditions to help prevent disease.** This form of cleaning uses special cleaners such as chlorine and anti-bacterial soaps to eliminate most bacteria and viruses. Sanitary cleaning and handling practices should be used in all environments where there is a chance of high numbers of bacteria and viruses.

People who work with food must wear protective gloves when they are handling food, to avoid spreading pathogens to people who will eat the food they serve.

Another reason they need to handle food carefully is to prevent food poisoning. Food poisoning results from eating food that is infected with harmful micro-organisms. Such food can make you very ill. In some cases, it can be fatal.

How Do You Get a Non-communicable Disease?

Communicable diseases are caused by bacteria and viruses spread by air, water, food, and contact.

Many non-communicable diseases are the result of bad environmental conditions, poor diets, unhealthy lifestyles, and genetics.

Check the figures that follow to see how each of these can result in disease.



Figure 9.7 Where do you find the most polluted air? Whether or not you smoke, it is probably in your home. This is because it is difficult to circulate fresh air inside a building. Exposure to polluted air (from cigarettes, for example) causes diseases such as emphysema. Emphysema is a lung condition in which breathing becomes difficult.

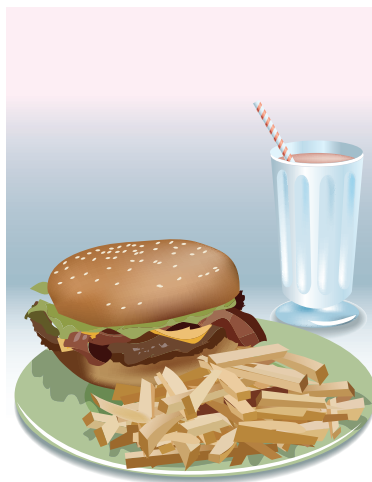


Figure 9.8 If you eat this meal once in a while, don't worry! If you eat this way regularly, however, you have a poor diet. People with poor diets often suffer from high levels of sickness. For example, a diet high in fat and cholesterol may cause heart disease. In heart disease, blood flow to the arteries supplying the heart becomes more and more restricted.



Figure 9.9 Screen time may be important to you. But you need to balance the amount of down time with active exercise. Lack of exercise can result in obesity. People who are 10 percent over the ideal body weight for their height and bone structure are obese. Being obese strains your heart's ability to function and increases your chances of developing Type II diabetes.



Figure 9.10 Genetics refers to the characteristics you inherit from your birth parents. Genetic factors are associated with diseases such as cystic fibrosis (CF), which damages the respiratory and digestive systems. Exercise may benefit people with CF by improving their strength and endurance.

Check Your Understanding

1. Define “communicable disease.” List three examples.
2. Define “non-communicable disease.” List three examples.
3. Make a table with three columns.
 - In the first column, list at least five diseases you or your family members have or have had.
 - In the second column, identify each disease as communicable or non-communicable.
 - In the third column, list the effects of each disease on health.

Key Terms

communicable
non-communicable
botulism
salmonellosis
sanitary

9.3 Who Is Most Likely to Get Sick?

In Canada, the average female can expect to live 81 years. The average male will likely see his 75th birthday. In Sierra Leone, an African country, the average female will live only 35 years and the average male 33 years. The difference has to do with living conditions.

Canada is a wealthy country. Most Canadians have enough food, clean water to drink, air that is relatively unpolluted, proper health care, and access to public education. The majority of adult Canadians also have paying jobs.

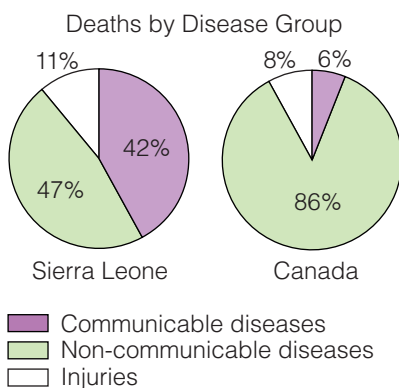


Figure 9.11 What fraction of people in Sierra Leone die from communicable diseases? How does that compare to the number of people in Canada who die from such diseases?

People in Sierra Leone have access to few or none of these things. In short, they live in **poverty**. This means that their basic needs for food, water, health care, and education are not met.

Children born in poor countries often die of starvation. They are also more likely to get a communicable disease. Because there is little health care, they often die from such illnesses.

Those who do survive to adulthood continue to be more likely to get a communicable disease. In Canada, pollution-control plants purify drinking water. In poor countries, drinking water comes directly from streams and rivers, which may be full of bacteria, viruses, and parasites.

DidYouKnow?

- More than one billion people lack clean water.
- More than two billion people lack toilets.
- Every year, nearly a billion people suffer from illnesses caused by contaminated water. The young and the elderly are the most at risk.
- Over half of the children that die every year die from malnutrition.

What Are Your Risk Factors?

No matter where you live, the same risk factors increase your chance of getting sick:

- not eating properly,
- drinking unclean water,
- not having a toilet,
- not practising personal hygiene, such as washing hands frequently,
- having sex,
- smoking tobacco, and
- consuming alcohol.

Table 9.2 Risk Factors and Disease

Risk Factor	Possible Disease	Source of Risk
not eating properly	obesity, diabetes, heart disease, cancer, tuberculosis	economic, social
drinking unclean water	diarrhea, parasites, <i>E. coli</i>	economic
not having a toilet	diarrhea, tuberculosis, parasites	economic
not practising personal hygiene such as washing hands frequently	tuberculosis, food poisoning	economic, social
having sex	AIDS and STDs	social
smoking tobacco	emphysema, bronchitis, and other respiratory conditions; arteriosclerosis and other circulatory diseases; cancer	social
consuming excess alcohol	liver disease, dementia, malnutrition	social



Figure 9.12 Natural disasters, like this flood in Badger, Newfoundland, in 2003, can result in sanitation problems. Once the ice thaws, water run-off provides breeding grounds for bacteria and viruses. As a result, victims may have limited access to clean water. Large numbers of injured people may overload medical facilities. People may get infections from the unsanitary conditions.

In poor countries, all of these problems are worse. In addition, food supplies may not be handled or stored properly. As a result, they may become contaminated.

Check Your Understanding

1. What is poverty?
2. Use Table 9.2 to identify what diseases you may be at risk of getting. Explain how you could reduce your risk of getting each one.
3. Defend the statement, “Access to safe water reflects the health of a country’s people.”
4. What type of disease kills most people in poor countries? in rich countries?

Key Terms

poverty

9.4 Stop Disease in Its Tracks

DidYouKnow?

Each year, there are a few cases of a form of food poisoning called **botulism**. The bacterium that causes botulism has the ability to grow in the absence of air and produce the deadliest toxin known. The death rate is high if it is not treated immediately. Botulism comes from Latin word *botulus*, which means sausage.

Today, dentists, first-aid volunteers, and even fast-food servers wear rubber gloves to protect themselves — and you — from disease.

Preventing Food Poisoning

The *staphylococcus* bacteria or **staph bacteria** that cause food poisoning enter food when you cough, sneeze, or handle food without washing your hands properly. Rubber gloves prevent some of the spread of these bacteria. Careful cleaning of the cooking utensils and area also help. Change dishcloths daily — bacteria grow in used dishcloths. Do not use sponges, which provide great breeding grounds for bacteria.

READING

Check

Choose five different methods of food preservation. Explain how each one prevents food poisoning.

Food Preservation

Preserving food prevents or limits the growth of pathogens that cause food poisoning. Table 9.3 explains how each method of food preservation helps prevent or limit this growth.

Table 9.3 Methods of Food Preservation

Method	Process and Effect on Decomposers
smoking or drying food	Removes moisture so that pathogens cannot grow.
canning	Jars filled with food are boiled in a hot water bath to kill existing pathogens.
salting	Removes moisture from cells and makes pH inhospitable for bacteria growth.
refrigeration	Bacteria need warm temperatures in order to live and reproduce. Refrigeration slows down the growth of pathogens.
freezing	- Freezing causes bacteria to enter a dormant stage. - It also prevents the growth of pathogens.
chemical additives	Additives such as sugar and salt slow the growth of bacteria.
vacuum packaging	- Bacteria need air in order to survive. Vacuum packaging removes air from the package. This reduces the number of bacteria that spoil food. - Unfortunately, some botulism bacteria like low-oxygen environments. They reproduce well in foods packaged this way, and can cause food poisoning.
irradiation	Food is exposed to certain types of radiation energy. This destroys pathogens in meat, fish, and poultry.
pasteurization	- Bacteria cannot survive in extremely high temperatures. Pasteurization involves heating a liquid such as milk to a high temperature for a specific period of time. The liquid is then cooled rapidly and refrigerated. - It destroys harmful bacteria without changing the flavour or nutritional value of the food.

Let's Go on a Picnic!



Consider the foods that have to be kept in a refrigerator. List 10 or more food products that need to be refrigerated to slow down the rate of food spoilage.

Think About It

Use what you have learned about safe food handling practices to plan a picnic for friends on a hot, sunny day.

What to Do

1 Use a chart to organize the following information.

- In the first column, list each of the foods on the menu: chicken wraps, coleslaw, home-canned black bean salsa, coconut cream pie, lemonade, fresh strawberries.
- In the second column, describe how it should be prepared.
- In the third column, describe how it should be kept safe from contamination.
- In the fourth column, identify what form of food poisoning might occur if it is handled improperly.

Analyze

1. (a) If the canned salsa is two years old and the seal is broken, what type of illness might the picnickers get?
(b) What would the symptoms be?
2. What advice would you give to a friend who wanted to eat potato salad left on a picnic table for several hours?

Extend Your Skills

3. Describe your favourite meal. For each food, list the steps you need to follow to keep it safe to eat. Include steps during and after preparation.
4. The picnickers decide to help clean up. If there is a choice between dishcloths and paper towels, which would you recommend using? Explain why.

DidYouKnow?

The glass dome over the salad bar is actually a sneeze guard to keep bacteria out of salads.



Figure 9.13



Figure 9.14 Florence Nightingale

Keeping Hospitals Healthy

Two hundred years ago, people expected to die when they went to the hospital. Hospitals were so unsanitary that patients often picked up diseases in the hospital. Today, this is less common because of the work of scientists such as Florence Nightingale, Joseph Lister, and Robert Koch.

In the 1800s, Florence Nightingale worked in a military hospital in Turkey. She reduced the death rate by making sure the hospital and patients were kept clean, improving water quality, and bandaging wounds to prevent infections from spreading to other patients.

Around the same time, an English surgeon named Joseph Lister realized that many patients died of infections following surgery. He began spraying carbolic acid on patient wounds and into the air of operating theatres. Carbolic acid kills pathogens on surfaces and on skin.

Robert Koch discovered that hot steam was effective in killing pathogens. He began using steam to destroy living organisms. This process is called **sterilization**.



What is the purpose of sterilization?

Aseptic Methods

Modern medicine uses many techniques to keep hospital rooms and equipment **aseptic** or **free of pathogens**. Aseptic practices include:

- using sterile bandages, instruments, linens, masks, and clothing during surgery;
- scrubbing all exposed skin of the surgeon before and after surgery;
- having surgical staff wear rubber gloves to prevent the spread of pathogens that might hide under fingernails.

In addition, antiseptics are used on people to disinfect the skin. Such antiseptics include iodine solution and alcohol.

Disinfectants are used on objects and surfaces to sterilize them. Some common disinfectants include iodine, chlorine, bleach, and antibacterial soaps.

Getting Ready...

- What is an epidemic and how does it affect society?
- How does your lifestyle affect your health?
- How does washing your hands stop the spread of disease?
- Where can you find information on diseases that might affect you in the future?
- What can you do to protect yourself from communicable diseases?



Londoners called 1858 “The Year of the Great Stink.”



How do you get rid of human waste? For centuries, people dumped it in water. By 1858, the Thames River in London, England was so full of sewage that the city stank. The smell was so strong that people in the House of Parliament could not work. Laws were passed to make sure the Thames River never got so bad again.

Who is responsible for preserving the quality of the water *you* use and drink?

The truth is that everyone has a certain amount of responsibility for keeping water safe to drink. The farmers in the second photograph

recognize that. That is why they are building a fence to keep cattle out of a stream.

In the past, many people died from diseases because they did not understand how diseases spread. In this chapter you will explore what people have learned about the spread of disease.

In the third photograph, researchers test water for possible contamination. In this chapter you will learn about other health programs and assess how important such programs are in building healthy communities.

to Disease



What You Will Learn

In this chapter you will learn:

- about widespread diseases and how they affect society
- about the importance of health programs and good personal hygiene
- how public health guidelines help maintain the health of a community
- how individuals help maintain personal and public health

Why It Is Important

- There are many public health services that can assist you in keeping healthy. But unless you know about them, understand their importance, and actually use them, they will not benefit you. Most of these programs are free — why not use them?

Skills You Will Use

In this chapter you will:

- list the effects of widespread diseases
- analyze the impact of contaminated water
- assess the need for public health guidelines
- assess the role of individuals in maintaining personal and public health
- design a public education campaign

Starting Point ACTIVITY

Responding to an Outbreak of Flu

Most people get the flu at some point. They suffer aching muscles, fever, chills, and congestion. Usually, these symptoms pass after a couple of days. But flu can be deadly. During 1918, over one billion people — half the world's total population — got the same flu. Many of them died.

Imagine you are an official responsible for your community's health. Use what you know about how diseases spread to design a poster showing ways people can protect themselves from the flu.

What You Need



coloured markers

poster board

What to Do

1. Brainstorm the content of your poster. To help you, make a chart with two columns.
 - In the first column, list factors that may contribute to the spread of flu.
 - In the second column, list how you might prevent the spread of flu.
2. Create a poster that shows people how to protect themselves from the flu. Use colour and visuals to attract attention to your poster.



10.1 How Diseases Affect Society



Figure 10.1 Three members of this family died within a month. Both historians and scientists read tombstones to learn about past diseases. But many members of one family dying from an epidemic do not tell the entire story. Consider how such deaths affect the community. For example, in farming communities, many deaths might result in fewer workers, more orphans, and food shortages.

European explorers brought many things to the New World — horses, iron pots, rifles, new seeds. Perhaps what affected Aboriginal groups the most were the diseases they brought — including smallpox, tuberculosis, flu.

Aboriginal groups were not able to fight the diseases because they had never before been exposed to certain illnesses. As a result, millions died.

Effect on Aboriginal Groups

The result of so many deaths was tragic.

- With fewer hunters, more people starved.
- Many young women died. The birth rate in Aboriginal communities slowed.
- Many elders died before passing on the knowledge of important traditions.

Pandemic Disease

Today, diseases such as AIDS and Ebola are affecting African society in a similar way. Despite modern knowledge of sanitation and personal hygiene, we are not able to prevent the spread of communicable diseases like AIDS because people continue doing things that increase their risk of getting such diseases.

Scientists have a special word for diseases, such as AIDS, that are extremely difficult to control. They refer to them as **pandemics**. Pandemic diseases affect a large proportion of the population over a large geographic area.

DidYouKnow?

Biological warfare is not a twentieth century invention. In the 1700s, a British general used disease as a weapon against Aboriginal groups. He distributed blankets from people who had died of smallpox. Aboriginals who used the blankets quickly contracted the disease and died, as did many others.

AIDS and the Law

Think About It

AIDS is a communicable disease that is spreading around the world. Some people infected with the AIDS virus are knowingly spreading it to others. Should these people be charged with a criminal offence? Use the following guide to help you develop and state an opinion.

What to Do

-  1 Use Internet sources to gather information about how AIDS is spread.
- 2 (a) Read and think about the articles that discuss whether or not people with AIDS should be charged with a criminal offence if they spread the virus.
(b) Discuss the ideas with your friends and family to get further input.
-  3 Make a decision about how you feel.
 - Do you think people should not be charged under any circumstances? Then say so and explain why. Remember to provide supporting points for your argument.
 - Do you think some people should be charged but not others? Explain who should be charged, and why. Explain who should not be charged, and why.
- Listen to other people's ideas.

Analyze

1. Think about the ideas expressed by others.
 - (a) List three points you might support. For each one, provide a reason for your support.
 - (b) List three points you reject. For each one, provide a reason for your lack of support.
2. Should people with AIDS be responsible for telling the people around them? Consider both sides by suggesting an argument for and against that each of the following people might make.
 - employer
 - teacher
 - sexual partner
 - member of the community
 - life insurance representative

Pandemic and Epidemic Diseases

Pandemic diseases affect a large proportion of the people over a large geographic area and are difficult to control. Contagious diseases that are not so widespread are called **epidemics**. This word is used when many people in a local area suffer from a disease at the same time.

Study the time line below to learn about outbreaks of various pandemic and epidemic diseases. Note how each of the outbreaks affected society. Also note the social conditions that contributed to the spread of each disease.

Figure 10.2

1346–1350 Bubonic Plague Pandemic

1500s to 1700s Tuberculosis Epidemic

The bubonic plague is a dangerous bacterial disease with symptoms of fever, chills, and swollen glands. From 1346 to 1350, it spread through Asia and Europe.

After the 1300s, the bubonic plague was not as widespread for several centuries. There were two more pandemics — one in 1542, the other in 1900.

Social Conditions

- overcrowded cities
- limited clean water
- no waste disposal facilities
- poor sanitation and hygiene
- high population of rats that carried the disease

Tuberculosis is a bacterial disease that damages the lungs. It was common in Europe and was spread around the world by European explorers.

In Canada, tuberculosis is responsible for a lot of illness and many deaths among Aboriginal peoples. Problems were particularly bad in the North. During the 1950s and 1960s, more than 70 percent of the Inuit in Keewatin were infected with the tuberculosis bacteria.

- Aboriginal population had no previous exposure to the disease
- overcrowding in boarding schools
- lack of sanitation
- poor diets

Impact

- 30 to 50 percent of the population died
- over 100 million deaths worldwide
- **People with the plague were put into quarantine.** They were held in a sealed-off area until they died or got better and no longer posed a disease threat to others. Only authorities with special permission could enter the quarantine area.
- crime rate dropped

- **patients were sent to special treatment facilities called sanatoriums**
- many children were put up for adoption instead of being returned to their families
- some Aboriginal adults lost contact with their communities

DidYouKnow?

In western Canada in the early 1900s, 25 percent of Aboriginal and Inuit children in residential schools died of tuberculosis. Aboriginal and Inuit people are seven times more likely to get this disease.

DidYouKnow?

The Spanish flu arrived in Alberta in October 1918, with 15 ill soldiers who were returning from World War I. Although the soldiers were isolated at Scarcee Military Camp, the disease spread. The numbers of infected people grew so high that the government banned public gatherings. More than 3000 Albertans died.

1918 Spanish Flu Pandemic

The Spanish flu is a viral disease that spread around the world during 1918. It killed thousands of people on every continent except Antarctica.

- soldiers travelling during World War I spread flu to people who had not previously been exposed
- poor sanitation
- poor diets
- little or no access to medical care as health-care professionals were treating sick and injured soldiers

- in Canada, more people died from Spanish flu (50 000) than from the war
- 21 million deaths worldwide
- loss of income for businesses such as theatres and restaurants because of fear of catching the flu
- crime rate dropped

1980s to present AIDS Pandemic

AIDS is a viral disease that weakens the ability of the body to fight diseases. AIDS was officially recognized as a disease in 1980. It is common around the world and increasing in severity. In Canada, there is approximately a 10 percent increase in new infections each year.

- sexual contact
- sharing needles (drugs, tattoos)
- infants born from infected mothers
- transfusions of infected blood

- fourth leading cause of death worldwide
- high costs for treatment; no cure
- fear and confusion about AIDS results in negative treatment of infected individuals

Social Conditions

Impact

DidYouKnow?

In some countries of Africa (e.g., Uganda), up to one-third of the people of child-bearing age are infected with HIV. As a result, many children are poorly cared for or are orphaned at a young age. So many people are unable to work that the economy has been adversely affected. The cost of treating AIDS is so high that governments cannot afford to treat victims.



Figure 10.4 Sanitary public toilets are not available everywhere around the world. Some countries use a cesspit, which is like an open hole. When the cesspit becomes full, urine and feces slop over the sides and could run into the street, spreading any disease pathogens that are present.

The Beginning of Public Health

Imagine living during the bubonic plague pandemic, when people did not know what caused disease. Imagine how frightened you would be.

At first, nothing was done to kill the rats that originally spread the disease. In addition, people dumped human waste in the streets. After some people began to understand that there might be a connection between sanitation and disease, city officials ordered streets to be cleaned and quarantined sick people in special facilities.

As people began to see the results of these kinds of measures, they became more interested in working together to develop healthy communities. This is referred to as **public health**.

Gradually, the government got involved in caring for public health. In England, “The Year of the Great Stink” helped speed the development of these laws. People who dumped waste and sewage in the Thames River were fined. Even though the new laws were not very well enforced, they were the beginnings of the public health system we have today.

Key Terms

pandemics
epidemics
quarantine
public health

Check Your Understanding

1. When does an epidemic become a pandemic?
2. Make a list of the impact of pandemic diseases on society.
3. List three pandemic diseases. Identify the social conditions that contributed to the spread of each one.
4. With a partner, use the Internet to research one of the following epidemic diseases: smallpox, cholera, typhoid fever, diphtheria, whooping cough. Develop a short presentation that mentions:
 - how it spreads
 - details about the disease
 - its impact on society

10.2 Improving Human Health

Health programs came to Canada after World War I as a direct result of the 1918 Spanish flu pandemic. Canadians realized they needed to rebuild the population. They also realized that they never wanted to suffer such a pandemic again.

In 1919, the Canadian government established its first department of health. Today, provincial health departments throughout the country do such jobs as:

- monitoring the safety of drinking water,
- checking the pollutants in the air we breathe, and
- taking measures to avoid outbreaks of disease.

In Alberta, the public health service is called the Regional Health Authority. This authority offers many public health programs and services. Study Figures 10.5 to 10.11 and the captions with each figure. As you do, make notes about the programs and services provided by public health organizations and how each improves human health.

Water Quality



Figure 10.5 Water quality specialists test beach water and report any problems to the public. For example, if the bacterial count in a lake reaches dangerous levels, beaches may be closed until the count returns to safe levels.

- Water is tested on a regular basis to ensure it is safe for drinking, cooking, bathing, or swimming.
- Water is treated to make it safe to drink. Treatment plants use chemicals such as chlorine to kill harmful bacteria.

Removal of Garbage



Figure 10.6 Most of our garbage ends up in landfill sites like the one shown here.

- Landfill sites prevent waste materials from harming the environment or entering the water system. Before the landfill is used, the ground is sealed to prevent toxic chemicals from accidentally entering the water supply.

Treatment of Human and Animal Wastes

- In sewage treatment plants, solid wastes are removed by settling or filtering. Helpful bacteria decompose any organic matter. The water is treated chemically to make it safe to drink.
- Large-scale livestock operations produce large quantities of manure that must be treated. Some producers store the manure in large lagoons on their farms. Modern treatment facilities separate solids from liquids, remove excess fertilizers, and recycle clean water for farm use. The liquid manure is then spread on cropland.

Air Quality

Air Quality Index (AQI) Categories			
0 – 15 Very Good	■	50 – 99 Poor	■
16 – 31 Good	■	100+ Very Poor	■
32 – 49 Moderate	■		

Figure 10.7 The Air Quality Index provides a reading of air quality and reasons for poor quality for many communities. Smog, which is a mixture of air pollutants, might be one reason for poor quality air. A smog alert warns people who are at risk or suffer from heart or lung problems to take protective action such as choosing to stay indoors.

- Air quality is tested and the results are provided to the public. Health Canada has developed standards and guidelines that list allowable levels of pollutants in the air you breathe. When the levels of pollutants are too high, people are advised to stay indoors and avoid heavy physical exercise. Physical exercise increases the body's need for oxygen, which may be difficult to get when there is poor air quality.



Describe four concerns that public health programs have addressed.

Mass Vaccination



Figure 10.8 When children are two months old, they begin receiving injections that provide them with protection from various contagious diseases. Most children get injections regularly until they are four to six years old. As they get older, they may receive booster shots to renew their protection.

- **Vaccinations** increase people's ability to fight off a disease. In Canada, children are often vaccinated against diphtheria, whooping cough, measles, mumps, tetanus, polio, and rubella. People who travel to other countries may get shots against yellow fever, typhoid fever, or hepatitis. They need to check health information for the destinations they are travelling to in order to adequately protect themselves.
- Public health officials often develop new vaccinations that respond to new outbreaks of disease. For example, meningitis is an inflammation of the covering of the brain. When this disease starts to spread in an area, public health officials provide free vaccinations against it to children and teens, who are most likely to get the infection.

Food Safety



Figure 10.9 Education campaigns reduce the number of illnesses spread by poor food handling. The campaign promotes four points:

- Clean — wash hands and surfaces often
 - Cook — cook to proper temperature
 - Separate — do not cross-contaminate (for example, keep separate cutting boards for meats and vegetables)
 - Chill — refrigerate promptly
-
- Health authorities use newspapers, radio, pamphlets, brochures, and the Internet to promote food safety awareness. Food inspectors check that workers in restaurants, food companies, and meat packing plants handle food safely.

The Choice Is Yours

These are just some of the programs developed to decrease the spread of disease and protect human health. The fact that you are studying this in school suggests that public health authorities consider educating people about health to be very important.

The next link in the chain is you. Are you concerned enough about your health to learn the many ways you can stay healthy? Do you care enough to maintain high standards of personal hygiene? When you get sick, are you willing to make sure that you do not spread the disease to others? Do you take time to learn how to protect your own and others' health and safety by taking training in first aid, swimming, lifesaving, or babysitter's training?

Check Your Understanding

1. In your own words, explain the role of public health in Canada.
2. Why is chlorine used to treat swimming pools and drinking water?
3. (a) How do high standards of personal hygiene affect individuals?
(b) How do such standards affect the community?
4. Describe three local public health programs. How does each one protect community health?

Key Terms

vaccinations

10.3 Public Health and You



Give three reasons why public health guidelines are necessary.

Vaccinations have reduced the number of people who die from communicable diseases. For example, smallpox has killed millions of people around the world. Today, the smallpox virus exists only in laboratories. Vaccination against smallpox is no longer recommended.

Does this mean that communicable diseases are no longer a serious problem? Should you just ignore public health guidelines?

The answer is no. There are many reasons for keeping an eye on public health guidelines.

Protecting Health

Diseases are not going away. In fact, many old diseases that were thought to have disappeared are resurfacing. These include polio and drug-resistant variations of tuberculosis. New diseases, such as AIDS and SARS, are also emerging. **Public health guidelines help to protect you from these and other diseases. Table 10.1 explains how.**

Table 10.1 Factors that spread disease and Public Health Guidelines

Factor	Concern	Public Health Guidelines
population growth and overcrowding	• overcrowded conditions increase the chances of being exposed to pathogens	• Good personal hygiene reduces the spread of pathogens. • Treating and disposing of wastes also reduces their spread.
migration	• diseases may be carried from country to country or from rural communities to cities	• Disease monitoring keeps an eye on the level of disease. • When levels are high, emergency response plans may prevent migration. • Travellers and their belongings are disinfected to prevent the spread of pathogens.
lifestyle	• people eat out	• Restaurant inspections ensure food-handling practices are safe and eating areas and utensils are kept sanitary.
travel	• increased exposure to disease (for example, eating improperly prepared foods, using unsanitary public washrooms)	• Travellers receive special vaccinations before travelling to some countries. • Travellers receive brochures explaining how to protect themselves from pathogens in food and water.
food supply	• food comes from many countries • food may harbour pathogens	• Food inspectors test foods for pathogens.
sexual activity and childbirth	• sexual contact results in spread of sexually transmitted diseases (STDs) • underweight babies, • unhealthy mothers	• STD awareness programs provide the facts. • Prenatal nutrition programs teach mothers proper nutrition. • Awareness education programs for fetal alcohol spectrum disorder reduce the number of mothers who drink during pregnancy.

Health Warnings

How do you know when there is a major health concern? Health authorities release warnings or advisories to alert you. These include:

- **food recalls** — In a food recall, all products containing a contaminated ingredient must be removed from store shelves. Radio and television news programs name the product, explain the danger, and tell the public what to do.
- **boil-water advisories** — When tests show drinking water is unsafe, health authorities warn members of the public to boil all water used for personal purposes such as drinking, cooking, and brushing teeth.
- **mass vaccinations** — When a communicable disease is beginning to spread, health officials provide free vaccinations for people likely to get the disease.
- **road safety** — In order to protect people who ride bicycles or drive vehicles, public health authorities provide information about the proper use of helmets, seatbelts, and other safety equipment. Alberta also has laws enforcing the use of child safety seats and bike helmets.



Figure 10.10 The concern about infectious diseases is worldwide. This Chinese poster tells people how to reduce the spread of scarlet fever. It says:

- Go to hospital as early as possible when infected.
- See a doctor as soon as you have a sore throat.
- Take the vaccination.



Figure 10.11 Public health officials encouraged legislators to pass a law requiring all cyclists under 18 to wear helmets when they are biking. Falling off a bicycle — even onto a soft surface when you are biking at a low speed — can cause brain damage.

Figure 10.12 Public health guidelines explain how to correctly install and adjust child car seats. The correct use of child safety seats protects infants and young children from injury.

From the Womb to the Grave

Public health guidelines help prevent disease and promote wellness. Study the visuals below and read the captions to see how public health programs can improve the quality of life during each stage of an individual's development.



Figure 10.13 The Canada Prenatal Nutrition Program educates pregnant women about lifestyle issues. This program works with women who have risk factors that may threaten their health and the development of their baby. For example, they work with young mothers to encourage them to eat foods that will help the baby develop properly. Similarly, they advise pregnant women that the use of alcohol, tobacco, or drugs negatively affects the development of a baby.



Figure 10.14 Children are vaccinated against common diseases before starting school. These vaccinations reduce the spread of communicable diseases. Vaccinations are provided for other high-risk groups too. For example, seniors, health-care workers, and teachers may get flu shots to protect them during the flu season.



Figure 10.15 Public health nurses work with teachers to promote health. For example, tobacco is an important public health issue. Using ads such as this one developed by Health Canada helps take the glamour out of smoking. Learning about health risks allows you to make informed choices.



Figure 10.16 Health guidelines promote staying active and fit to live long and live well. Practising Taoist Tai Chi is one way that you can keep fit mentally and physically.

READING
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What are two ways in which public health services help prevent disease?

How Important Are Your Personal Actions?

Proper hand washing is only one aspect of personal health. You have other important choices to make. Consider these.

1. Protect your own health — take care of your personal hygiene; get vaccinations; keep informed about health advisories
2. Make wise lifestyle choices — exercise regularly; eat a well-balanced diet; avoid smoking, alcohol, and non-prescription drugs; avoid sexual contact
3. Adopt safer practices — wash your hands often; wear bike helmets and seat belts; when you become sexually active, practise safe sex

Your choices — wise or otherwise — will affect the people around you.



How do personal actions contribute to the health of your community?



Figure 10.17 How does washing your hands protect others from disease?

Check Your Understanding

1. List five factors that account for the spread of communicable disease.
2. Do you think it is important to have public health guidelines? Use examples to help you explain why or why not.
3. Defend the statement, “An ounce of prevention is worth a pound of cure.”
4. What are three steps you can take to protect your own health? Provide a specific example of each.

Protecting Yourself

Getting Ready...

- How does your body protect itself from disease?
- What does your immune system do?
- What vaccinations have you had?
- How might using medicines improperly harm you?



Don't touch the keyboard! There's a chicken pox virus hanging around.

Hold your breath — someone just sneezed!

You have many opportunities to be exposed to disease. Why don't you get sick more often?

The answer is that your body has several lines of defence against disease. In this chapter you will learn about the ways your body protects itself from invaders. You will learn about treatments available for when you *do* get sick. For example, you can prevent tetanus, which is caused by

bacteria that enter your body through a cut, by getting a vaccination against it, and then continuing to get booster shots periodically.

Are your vaccinations and booster shots up to date? By the end of this chapter, you might want to check.

Take a look at the mould in the photograph at the top of page 193. Penicillin, which was the first antibiotic developed, comes from this mould, which grows on bread! Today, there are many types of antibiotics. In this chapter you will also learn why you need to use antibiotics properly.

from Disease



What You Will Learn

In this chapter you will learn:

- how your body's physical defences prevent infection
- how your blood helps control pathogens
- how your immune system protects you
- how antibiotics and vaccinations prevent or treat disease
- about the dangers of using antibiotics improperly

Why It Is Important

- You have likely taken antibiotics in order to fight infections. When antibiotics are used incorrectly, the bacteria and viruses in your body become resistant to them and the antibiotics may no longer work. Knowing how this happens will help you use antibiotics so they continue to be effective.

Skills You Will Use

In this chapter you will:

- identify physical defences against infection
- explain the role of blood in controlling pathogens
- describe the body's immune system
- describe how antibiotics and vaccinations prevent or treat disease
- analyze cold remedies

Defences against Pathogens

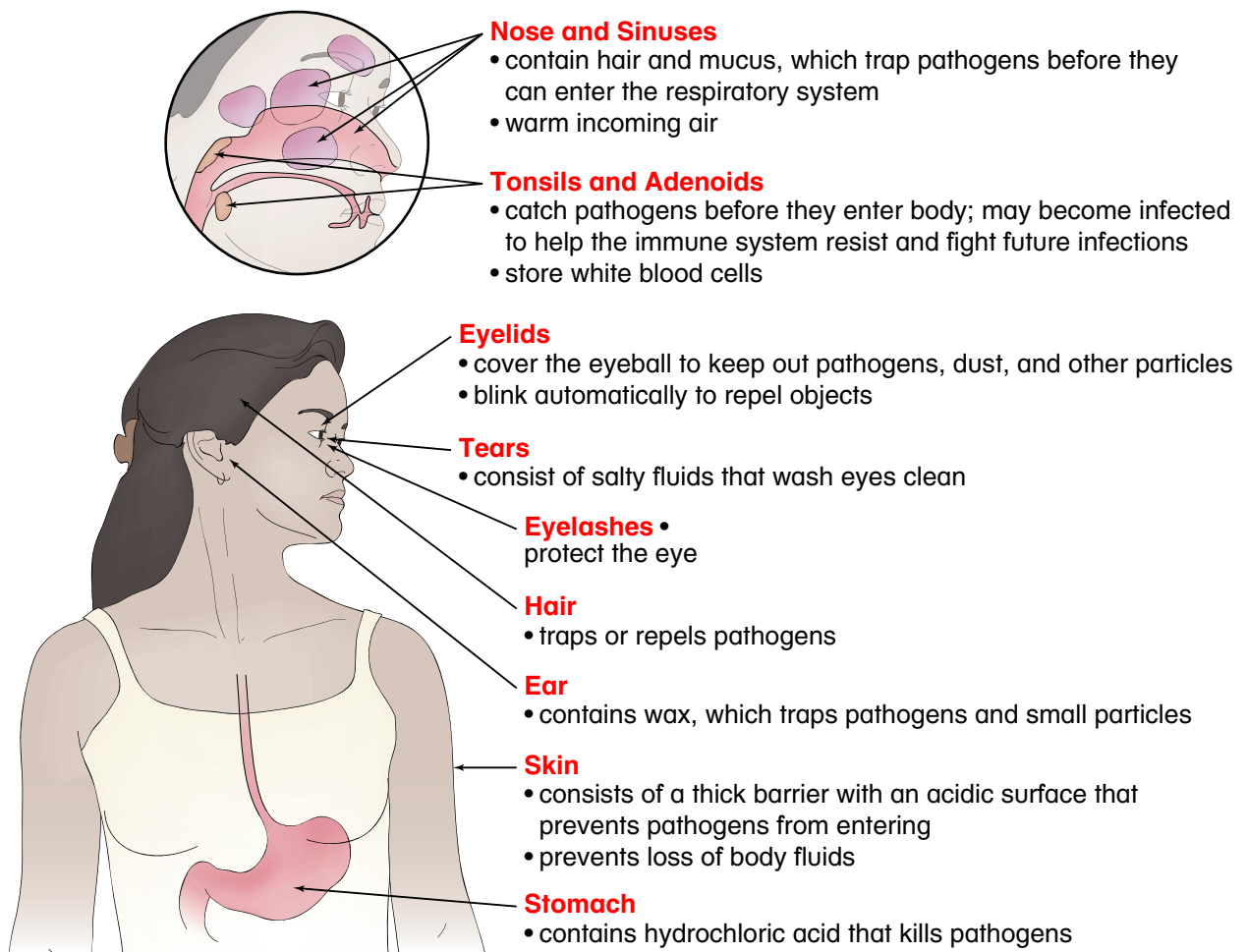
READING check

List the body's physical defences against disease.

Millions of potentially harmful pathogens surround you every day. They live on your body and on most surfaces you touch. So why aren't you sick all the time? **It is because your body has three lines of defence against pathogens: physical defences, the inflammatory response, and the action of antibodies.** Antibodies are released by white blood cells in order to destroy pathogens that have invaded the body.

Figure 11.1 shows you the **physical defences**. This first line of defence involves body structures such as the skin and secretions such as mucus. Each of these forms a physical barrier against pathogens. Read the labels with Figure 11.1 to learn how your body's physical defences work and where they are located.

Figure 11.1 Physical defences



The Inflammatory Response

Have you ever got a splinter — and soon afterward noticed an inflammation around the area? This is an example of an **inflammatory response**. During an inflammatory response, the blood vessels surrounding the injury swell. The area becomes red and warm as more blood arrives.

The body has several inflammatory responses.

- If pathogens get past the physical barrier created by the skin, the body sends white blood cells, that are apart of the immune system, to repel the invader. Study Figure 11.2 and read the caption to see how this works.
- An inflammatory response can also occur when the skin is hit, for instance, when you fall and hit the ground. This response is caused because blood oozes from the ruptured blood vessels into surrounding tissue. White blood cells engulf damaged cells and cell parts at the site of the bruise in order to clean up and heal the bruised area.
- Any kind of infection (e.g. tonsillitis or appendicitis) will trigger an inflammatory response. The ‘itis’ ending of these words means “inflammation of.”

Physical defences and the inflammatory responses are called general defences because the body reacts in the same way regardless of what kind of pathogen is invading. For example, you will blink when something enters your eye. This happens both when it is something you want there, such as a contact lens, or something damaging, such as dust.

The third line of defence is the formation of antibodies in the **immune system**. This line of defence targets specific pathogens and will be discussed in detail in the next section of this chapter.

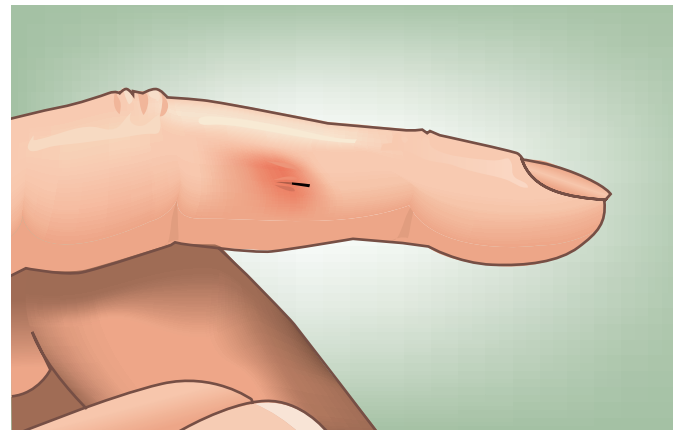


Figure 11.2 The blood vessels surrounding this splinter swell. The area becomes red and warm as more blood arrives. Specialized white blood cells called **macrophages** surround and destroy damaged cells and pathogens such as viruses and bacteria. Macrophages engulf the splinter and start destroying it.

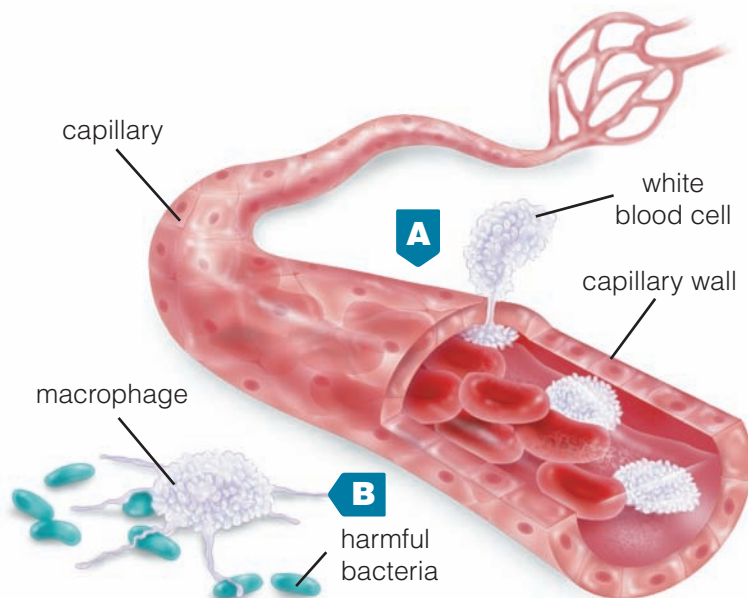


Figure 11.3 Look at how macrophages engulf invading cells. White blood cells leave capillaries (A) and engulf harmful bacteria (B). Macrophages can consume large numbers of pathogens at once.

11.2 Immunity

Have you or someone you know ever had the chicken pox? Few of your friends may have had this disease, since there is now a vaccination for it. However, even if you have not been vaccinated, if you have already had the disease, you are unlikely to get it again. This is because you have built up immunity — or resistance — to chicken pox. If you were exposed to the virus again, your immune system would recognize and resist it.

Your body recognizes substances called **antigens** that are not part of itself. Antigens are specifically shaped structures that are attached to pathogens. All living tissues such as viruses, bacteria, and fungi contain antigens. So do pollen, transfused blood, transplanted organs, and cancer cells.

How does your immune system respond to the antigens it encounters? Most of this is done at the cellular level by white blood cells. White blood cells make and release Y-shaped protein compounds called **antibodies**. Antibodies bind to specific antigens. This makes it easier for white blood cells to destroy them.

- When you first contacted the chicken pox virus, your body did not have antibodies against the chicken pox antigen. It immediately started producing them.
- Once your body has antibodies to a pathogen such as chicken pox, your immune system recognizes and destroys the antigen whenever it shows up again. This is why you are considered to be immune to the disease.

Figure 11.4 explains how antibodies combine with antigens.

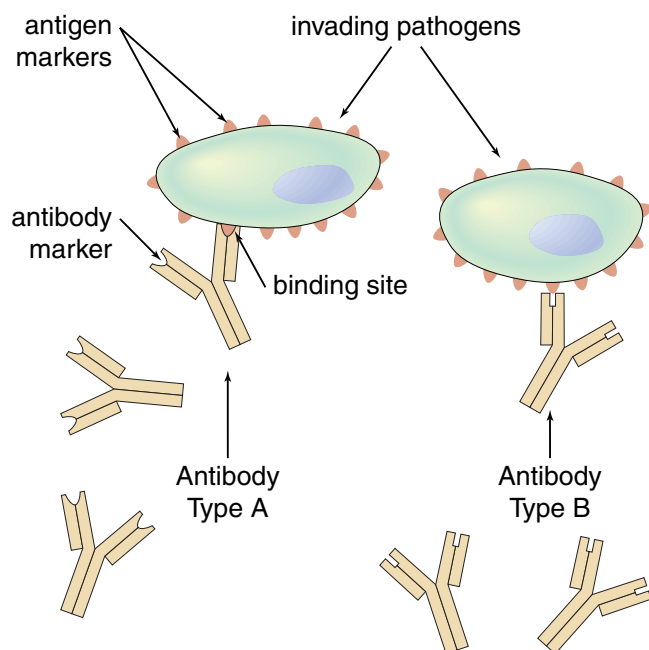


Figure 11.4 Each antibody has a shape that matches a specific antigen. This means that an antibody produced against a cold virus, for example, would not fit with an HIV antigen.

- Antibody Type A attaches one type of antigen with a complementary shape, like a lock and key. For example, a flu antibody would combine with flu antigens.
- Antibody Type B does not combine with the antigen because their shapes do not match. Notice the outer edge of each arm of the antibody and the antigen. The antibody markers are different from those found on the antigen.

DidYouKnow?

The average person has more than 100 000 antibodies waiting for the invasion of antigens.

READING check

Describe how antigens and antibodies interact.

Types of Immunity

You are born with a certain amount of immunity. This is called **inherited immunity**. Inherited immunity refers to the resistance that organisms have to most diseases. This form of resistance is passed on from your mother.

From birth on, you build what is called **acquired immunity**. Figure 11.6 shows that acquired immunity is divided into two types according to how you get it.

- Your body develops **active immunity** by making antibodies against diseases to which you have been exposed. **This is the type of immunity you might develop against chicken pox.**
- You acquire **passive immunity** from antibodies produced in another animal and put in your body. **This is the kind of immunity you get when you get a vaccination.**

READING Check ✓

Explain the different ways that people can develop or acquire immunity. Give an example of each.

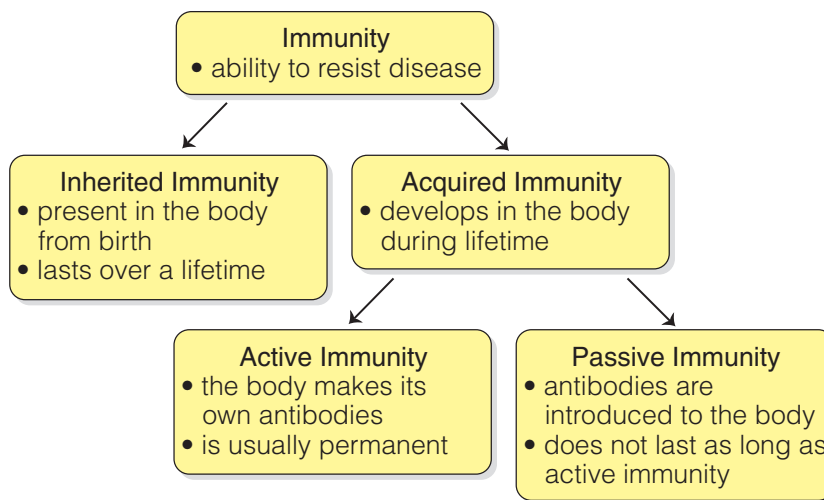


Figure 11.6



A. Naturally Acquired Active Immunity "Strawberry tongue" is often a symptom of scarlet fever. Once you have had scarlet fever, you are unlikely to get it again. That is because your body builds immunity against the disease.



B. Artificially Acquired Active Immunity Hepatitis B vaccine works well against hepatitis in teens. This provides artificial immunity. The hepatitis virus is cultivated by infecting yeast cells.



C. Naturally Acquired Passive Immunity As long as babies continue to breast-feed, they are immune to many of the same diseases as their mother.



D. Artificially Acquired Passive Immunity Small amounts of snake venom are injected into animals such as cows or horses to create antibodies. The antibodies are called antitoxins. The antitoxin is removed from the animal's blood and used to provide immunity in people who are bitten by that type of poisonous snake.

Blood Transfusions

Do you know which blood type you are?

There are four blood types: A, B, AB, and O. Doctors check patients' blood type before giving a transfusion. That is because the blood type of the transfusion must match the blood type of the person who gets it.

Blood types must match because there are two kinds of antigens on the red blood cells of humans and two kinds of antibodies in the plasma. If the blood types do not match, antigens will react with antibodies, causing blood cells to clump, clumping can lead to heart attacks and strokes.

Following is a chart that outlines donor blood compatability.

Recipient Blood Group O	Recipient Blood Group A	Recipient Blood Group B	Recipient Blood Group AB
Donor Can be	Donor Can be	Donor Can be	Donor Can be
O	O or A	O or B	O, A, B, or AB

11.3 Immunization

Why do some people get a flu shot each year?

They know that their body can develop immunity by getting either an injection of **vaccine**.

- A vaccine consists of weakened or dead disease pathogens. Once a vaccination or **immunization** is inside your body, your body begins to produce antibodies against the antigens in the vaccine. The disease is weaker than usual because of the weakened or dead pathogens.
As a result, you get few or no symptoms. After your body produces antibodies, you have lasting protection against the real disease.

Key Terms

antigens
inherited immunity
acquired immunity
active immunity
passive immunity

Why Get Immunization Shots?

Some people do not get immunized for personal or medical reasons. For example, some people react to certain immunizations. Such reactions often occur in people whose immune systems have been weakened by a disease such as asthma or by severe allergies. In such people, whooping cough vaccine can occasionally cause reactions such as high fevers or seizures.

If vaccinations can cause such reactions, why do most people get them?

Basically, immunizations prevent more problems than they create. Immunizations have reduced the incidence of communicable diseases and the number of deaths from them. Worldwide immunization against smallpox, for example, has actually eliminated that disease.

Recently, some countries have stopped or limited immunization programs. As a result, there have been increases in diseases, such as polio, that had almost disappeared.



Why should you get immunizations?

Allergies

Allergies develop when the immune system overreacts and becomes sensitive to substances such as pollen, dust, or mould. The immune system then treats the substance as though it were a pathogen and begins to build antibodies against it. These antibodies result in a variety of symptoms. Mild reactions include a runny nose and itchy or watery eyes. Severe symptoms can include breathing difficulties, and can build to extreme allergic reactions that cause death.

Scratch tests identify allergens. Drops of antigen are absorbed under the scratched skin. Allergic reactions produce redness, swelling, and itching along the scratch.

People who suffer from allergies sometimes get immunotherapy. This involves injections of weakened allergens. Regular injections build the number of antibodies against the allergen.

See Table 11.1 for the list of preventable diseases most Canadians are immunized against, and the effects of each disease.

Disease	Effects of Disease
diphtheria	- infection in nose and throat can make breathing difficult - skin infections, rapid pulse - heart failure, paralysis, death
measles	- red rash, high fever, cough, sore throat - often ear infections and pneumonia
mumps	painful, swollen salivary glands; fever; headache
polio	- fever, headache, stiff neck and back, muscle weakness - often permanent paralysis, death
rubella	- mild fever, rash, swelling of the neck - dangerous during early pregnancy; causes miscarriages and handicaps
tetanus	- muscle spasms, difficulties breathing - death
whooping cough	- severe coughing, difficulties breathing - often pneumonia - sometimes brain damage and death

Table 11.1 Preventable Diseases
You were born with all of your mother's antibodies. These antibodies lasted a short time, longer if you were breastfed. Childhood vaccinations allow you to continue having the benefits of this immunity.

11.4 Treating Infectious Diseases

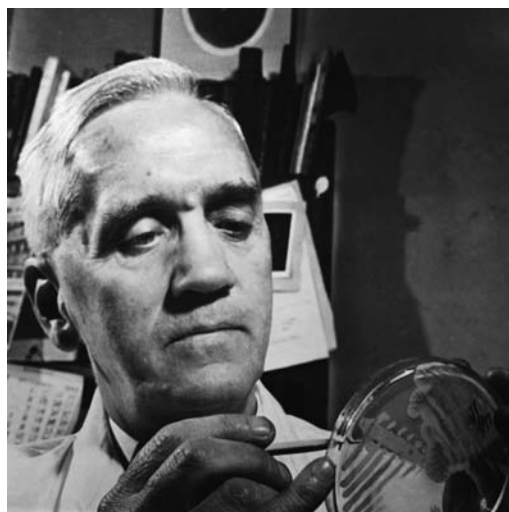
READING check

How can you decide between using an antibiotic or an over-the-counter drug?

No vaccine has yet been developed that can prevent colds! That's because the cold virus keeps mutating. As fast as someone tries to develop a vaccine for the common cold, a new cold virus appears. When you get a cold, your body is usually able to heal itself within about a week. There are times, however, when your body needs help.

Usually, people try **over-the-counter drugs** to relieve or treat symptoms. Over-the-counter drugs are sold without a doctor's prescription. They relieve symptoms such as fevers and headaches that accompany colds, but do not kill the viruses that cause the cold.

Figure 11.8 A blue mould called *penicillium notatum* became famous after Sir Alexander Fleming of Scotland discovered it by accident in 1929. He noticed some mould colonies growing in a dish of bacteria he was studying. No bacteria were growing around each mould colony. Fleming concluded that the mould produced a substance that killed the bacteria. His work led to the discovery of penicillin — the first antibiotic. Today scientists develop antibiotics from fungi, moulds, and bacteria.



If symptoms of infection continue, your doctor may prescribe an **antibiotic**. Antibiotics fight bacterial infections by either killing bacteria or preventing them from reproducing. Each antibiotic is effective against specific types of bacteria. They do not, however, help with a viral infection.

When treating a cold or strep throat, you need to know whether or not the infection is bacterial or viral. Viruses cause colds, the flu, most sore throats,

coughs, and sinus infections. Since antibiotics do not work against viruses, the best thing you can do for these is to drink plenty of fluids, get rest, and perhaps take over-the-counter drugs to relieve any symptoms. Since bacteria cause strep throat and pneumonia, an antibiotic might be effective in treating these.

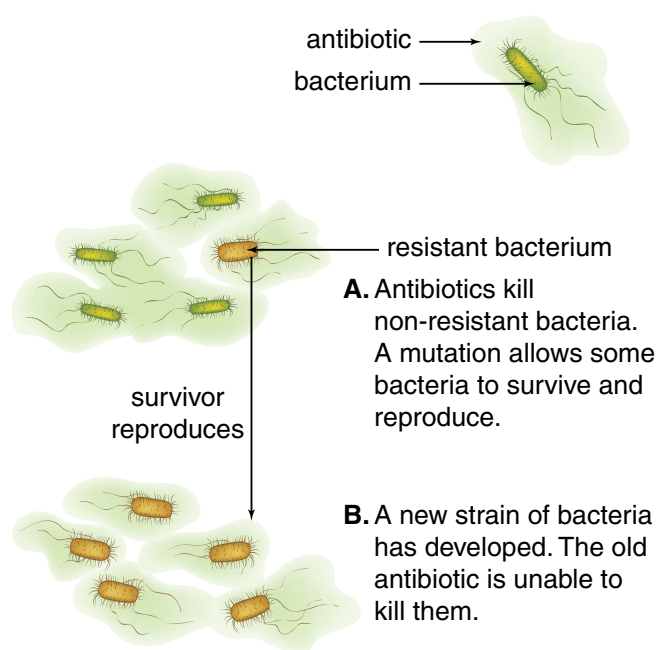


Figure 11.9 Scientists test antibiotics by growing bacterial colonies and introducing small disks of antibiotics to the bacteria. You can see the mould is killing some bacteria. Scientists determine the effectiveness of antibiotics by measuring the size of the “no-growth” zone.

Problems with Antibiotics

Some people demand antibiotics to fight every cold and sore throat they have. This is not a good idea for the following reasons:

- **They kill harmful and helpful bacteria** — Your body contains many helpful bacteria that aid in such things as digestion. Antibiotics kill these, and can cause symptoms such as an upset stomach, diarrhea, or vaginal yeast infection.
- **They can cause side effects** — Some people develop allergies to antibiotics. Report any symptoms, such as an itchy rash, to your doctor.
- **Overuse can help the development of resistant bacteria** — Figure 11.10 shows how bacteria can build resistance against an antibiotic. Study this figure and the accompanying caption.



READING
check ✓

Explain how resistant bacteria develop and what you can do to help.

Figure 11.10 When you take an antibiotic, it usually kills bacteria over a period of seven to 10 days. Regularly occurring mutations mean that some of the bacteria are immune to the antibiotic. These bacteria survive and reproduce, forming a new antibiotic-resistant strain of infection. Not taking all of the antibiotic the doctor prescribes can increase the chance of bacteria becoming resistant.

The resistant bacteria are often called superbugs. When the superbugs reproduce, the new strain of the disease begins to infect many people. People die from infections that used to be treatable because the new superbugs are resistant to one — or sometimes many — antibiotics.

Even if you are a healthy person who has never taken an antibiotic, you could become infected with a superbug from other people. That is why bacterial resistance poses such a threat. The best way you can help avoid this is by using antibiotics only when necessary and following the doctor's directions for their use.

DidYouKnow?

Drug reactions account for one in five hospital admissions. Nine out of 10 patients do not take medicines properly. For example, combining medicines may produce effects different from those produced when a medicine is taken alone, or it may produce side effects.

Use Antibiotics Safely

- **Ask questions** — Ask your doctor or pharmacist about precautions to take when using a medicine, and about any side effects. Before you take two or more medicines at the same time, ask if they can be taken together safely.
- **Read the label** — Find out how and when to take medicine. Some medicines must be taken with food; others on an empty stomach. Some medicines cause sensitivity to the Sun. Also check the expiration date and discard any medication that has passed the expiry date.
- **Take drugs properly** — Follow the directions on the label until you have taken all of the medication. Not doing so may lead to the development of antibiotic-resistant bacteria.

Handling Over-the-Counter Drugs

In most cases, over-the-counter drugs are considered safe to use. However, there are risks if you treat yourself without professional advice.

Generally, you should not assume that you have a specific disease and choose a medication to treat it. The same symptom can result from one of several diseases, and you may be treating the wrong ailment. For this reason, avoid self-diagnosis unless you are dealing with a problem you have had before and have exactly the same symptoms.



Figure 11.11 Many different diseases have the same symptoms. For this reason, it is best to check with a doctor or pharmacist before buying an over-the-counter drug.

Choosing Over-the-Counter Drugs

When you have a choice among several medications, read the labels carefully.

- Find out what symptoms the medicine treats. If you have a headache, look for something that relieves headaches.
- Read the warnings and list of side effects. Many medications make you drowsy. If you must take one of these, you may need to ask a friend to drive or postpone a date.
- Use exactly the amount of medication prescribed, and use it for the recommended length of time.



What information should you read on the label of an over-the-counter drug?

Getting Ready...

- What characteristics have been passed from one generation to another in your family?
- What environmental factors might affect the development of a fetus?
- What determines whether a baby will be a girl or a boy?
- How does overexposure to the Sun affect cells?
- What is the Human Genome Project?



Can you roll your tongue? Are your earlobes detached? Can you bend back your thumb? These are just a few of the characteristics that you may have inherited. In fact, millions of combinations of characteristics are passed from one generation of a family to the next. Study the photograph of the extended family. What characteristics do you think the grandparents passed to the children?

People can also inherit certain disorders. In this chapter you will learn more about heredity and some inherited disorders.

Recently, scientists identified all of the characteristics that make up human

beings. Now they are studying diseases caused by alterations in genetic make-up. You will investigate some issues related to these advances in research.

The environment plays a significant role in human development and health. Exposure to environmental elements may cause changes in inherited characteristics. For example, what an expectant mother eats, drinks, or breathes may affect her unborn child. You will investigate the effects of environmental factors on health.

As you learn about heredity and see what a complex combination of traits you have, you may decide to trace some traits in your own family.

Health



What You Will Learn

In this chapter you will learn:

- how physical characteristics are inherited in humans
- how chromosomes determine the sex of a child
- how genetic information is organized in the cells of your body
- about human diseases that result from changes to genetic make-up
- about ethical and social issues related to genetic research

Why It Is Important

- Some babies are born with disorders that could have been prevented. Identifying the risk factors that may cause preventable disorders can help you protect your health and the health of children born in the future. This is important for both males and females, since both provide genetic material to the fetus.

Skills You Will Use

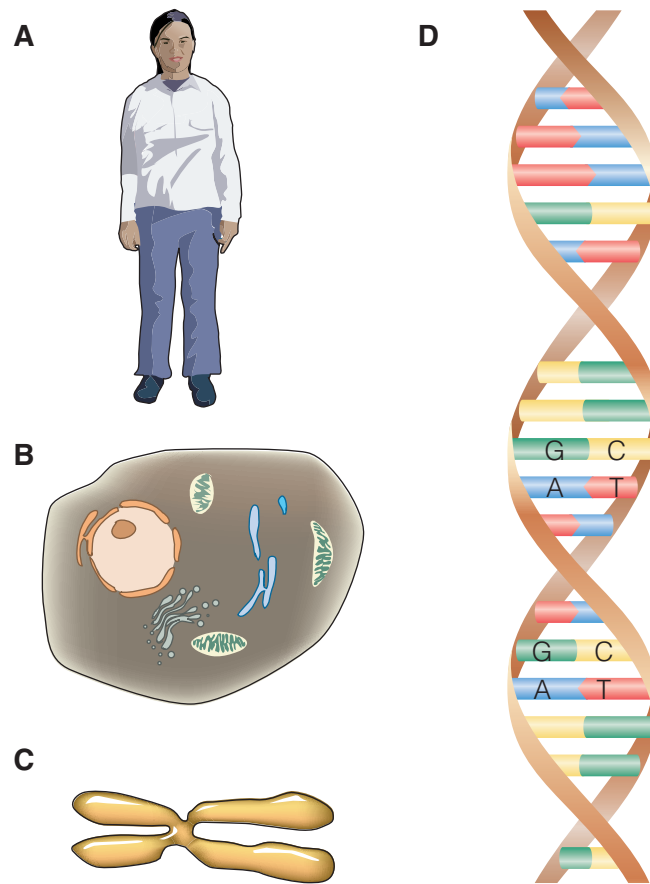
In this chapter you will:

- use techniques to illustrate and interpret hereditary characteristics
- predict the occurrence of some characteristics passed from parents to offspring
- construct a model of how genetic information is organized in cells
- investigate the effects of changes in genetic make-up
- identify points of view about issues related to genetic research

12.1 Structure of DNA

Figure 12.1

- A.** Your body contains about 10 trillion cells.
- B.** Each cell contains a nucleus with **chromosomes**.
- C.** Each chromosome consists of double-helix shaped molecules of DNA. This chromosome has an identical copy attached because the cell has just divided.
- D.** DNA is divided into segments called **genes**. These control or determine inherited traits.



People may have told you that you look like your mother or dad — or perhaps a grandparent. Do you know how these characteristics get passed down to you?

The answer involves a branch of science called **genetics**. **Genetics is the study of how characteristics or traits are passed from parents to offspring.**

To understand how you ended up with the traits you did, think about how your body is made up of systems, organs, tissues, and cells. The unique code for your genetic make-up is stored in the nucleus of each one of these cells. Figure 12.1 explains how your genetic information is organized. Study the four parts of the diagram.



Draw a picture to show how DNA is organized.

The Structure of DNA

Look closely at how the DNA is organized in part D of Figure 12.1. Notice that it appears as a twisted ladder. Look at the rungs on the ladder.

Chemical compounds called bases make up the rungs. A pair of bases forms each rung. One molecule of DNA contains about three billion base pairs.

Different genes are made up of different arrangements of these base pairs. The sequence of the base pairs determines the traits you inherit.

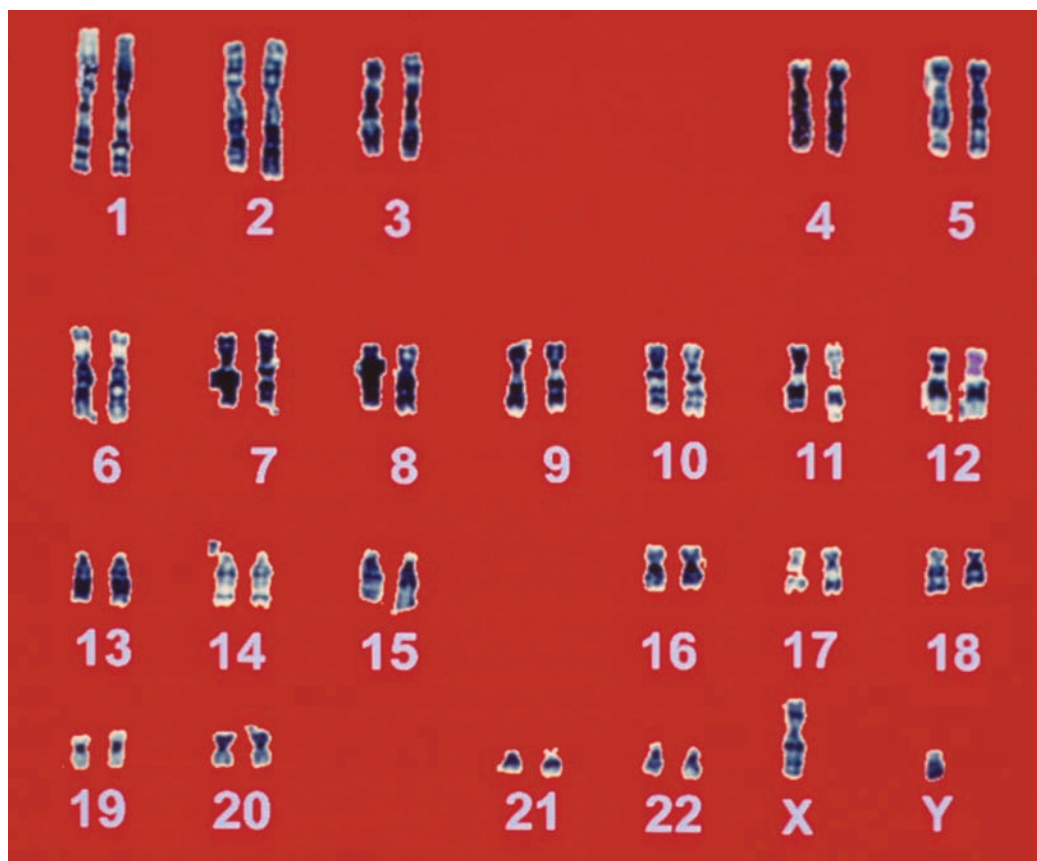
Chromosomes

Human body cells contain 46 chromosomes arranged in 23 pairs that hold all the genetic information. One chromosome of each pair comes from the mother and the other chromosome of each pair comes from the father.

Figure 12.2 describes how scientists study chromosomes. First, they group them into pairs, and then they organize them by size, beginning with the longest pair. The sex chromosome pair is organized last. **In humans, there are two kinds of sex chromosomes: X and Y. The Y chromosome is found only in males. Females have XX chromosomes. Males have XY chromosomes.**

READING
check ✓

What chromosome combination determines the sex of human offspring?



DidYouKnow?

The DNA of all people on Earth is 99.9 percent identical. It is the 0.1 percent difference that makes each one of us unique.

Figure 12.2 Scientists put a sample of white blood cells that are dividing under a microscope and take a photograph. They use the photograph to make a chromosome map like the one shown here. Note that the chromosomes are grouped in numbered pairs according to their size and shape.

Find the pair of sex chromosomes. How large is the X chromosome compared to the Y? Does this map represent a male or a female? How do you know?

12.2 Inheritance and Genetics

Half of your genetic make-up comes from your mother and the other half from your father. Then why don't all brothers and sisters look alike? How do traits skip generations?

The answers to these questions have to do with **heredity**, which is the passing of characteristics from parents to offspring.

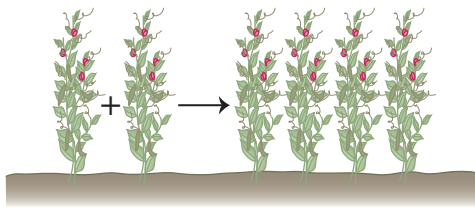


What is heredity?

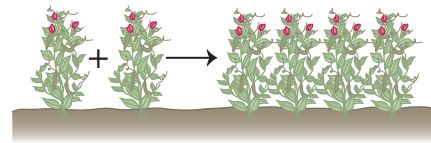
During the mid-1800s, Gregor Mendel, an Austrian monk, experimented with pea plants to show that traits do not happen by accident. He noticed that pea plants vary in height. Some are tall, others short. He crossed tall and short pea plants to see how height would be affected in the offspring.

Mendel recorded the traits of the offspring plants over several generations. Figure 12.3 summarizes what he discovered about the trait of tallness in pea plants. Study the pictures and read the captions to help you understand the difference between dominant and recessive traits.

Figure 12.3

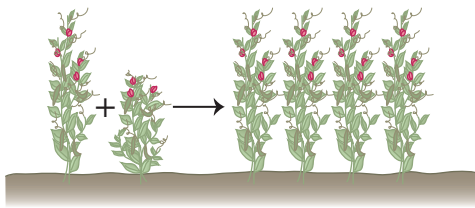


A. tall parent + tall parent = pure tall plants

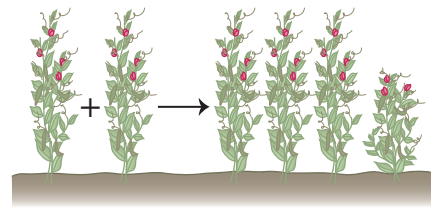


B. short parent + short parent = pure short plants

• Mendel looked at the results of the crosses shown in A and B. He referred to the parent plants as purebreds. **Purebreds** have identical genes for each trait. In this case, the parents in A and B had identical genes for tallness.



- C. pure tall parent + pure short parent = hybrid tall plants
- When Mendel crossed the pure tall plants with the pure short plants, the offspring were always tall. The short trait seemed to disappear.
 - Mendel called tallness a **dominant** trait because it seemed to dominate the short trait.
 - He called shortness a **recessive** trait because it seemed to disappear.
 - Offspring of this cross are called hybrids. **Hybrids** carry one gene of each kind for each trait. In this case, they carry one gene of each kind for tallness.



- D. hybrid tall plant + hybrid tall plant = mix of tall and short plants (3 tall, 1 short)
- Mendel crossed hybrid tall plants with hybrid tall plants. He was surprised to discover that sometimes shortness appeared in the offspring.
 - From these observations, Mendel concluded that tallness must be controlled by factors inherited from both parents.

Recording Inherited Traits

Since chromosomes come in pairs, genes for traits do too. Many traits are controlled by only one pair of genes, others by two or more pairs.

For traits controlled by one pair of genes, geneticists use two letters to represent the **genotype** or combination of genes.

- Capital letters represent dominant genes. For example, the letter T could represent the dominant gene for tallness, which is the *tall* gene.
- Lower case letter represent recessive genes. For example, the letter t could represent the recessive gene for tallness.

During reproduction, chromosomes separate so that each sperm or egg cell gets only one chromosome of each kind. As a result, only one gene for each trait is passed on to the offspring from each parent.

Punnett Squares

Geneticists use a **Punnett square** to predict the probability of offspring inheriting certain traits. Figure 12.5 shows the results of crosses in Figure 12.3.

A			B			C			D		
											
	TT	TT		tt	tt		Tt	Tt		TT	Tt
	TT	TT		tt	tt		Tt	Tt		Tt	tt

Figure 12.5 Describe the appearance of each offspring in A, B, C, and D. Why were there no short plants in C, even though one plant was short?

Note that:

- the possible genes in the eggs from the female are across the top;
- the possible genes in the sperm from the male are down the left side; and
- the possible gene combinations of offspring are shown in the four squares.

There are four possible gene combinations in the offspring. Look at the offspring in Punnett squares C and D. Mendel used results like these to show that recessive traits can skip generations.

In Punnett square C, shortness did not appear in the offspring because tallness is dominant. However, shortness reappeared in the next generation (Punnett square D). This occurred because one of the offspring inherited two genes for shortness.

Try This!

Females have XX genotype. Males have XY genotype. Complete a Punnett square to discover the chances of giving birth to a male or female child.



Figure 12.4 This lowland gorilla has no pigment or natural colouring of its skin, eyes, or hair. Such an individual is called an albino. Albinos have inherited two recessive genes. They have two copies of the gene for albinism and no copy of the gene that makes normal pigment.

A type of scientist called a geneticist uses a tool called a **family tree or pedigree** to trace traits in a family. A pedigree is a diagram that shows the history of a trait from generation to generation. Pedigrees show traits with different characteristics, such as long eyelashes or short eyelashes, through many generations. Plant and animal breeders use the information from pedigrees to increase the chances of breeding plants and animals with only desirable traits.

12.3 Genetic Changes

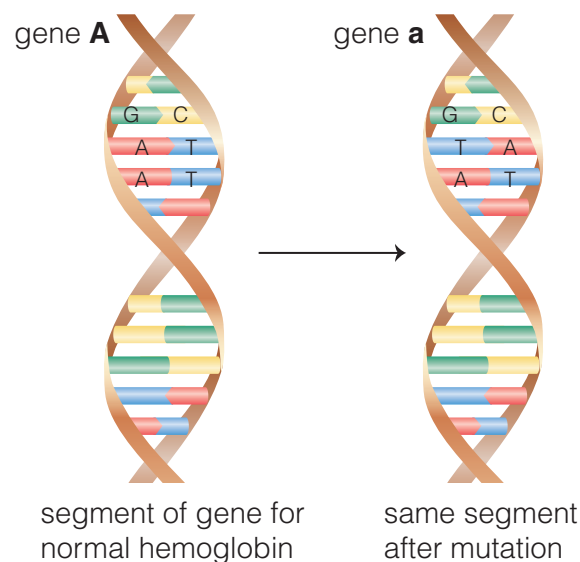
READING check

Explain how inherited and environmental mutations can occur.

Most of the time the process of cell reproduction and copying genes works well. Mistakes, however, can happen during reproduction when chromosomes are copied. In most cases, body cells recognize errors and fix them. In other cases, a cell's repair mechanisms can fail. These mistakes are then passed from generation to generation.

Such changes in genetic information can result in an abnormal gene called a **mutation**. Mutations can occur in several ways. Figure 12.7 shows how the change in one of the base pairs of a gene can lead to sickle-cell anemia.

Figure 12.7 Look at the pairs in gene **A** and gene **a**. How do they differ? Gene **A** produces normal hemoglobin. Gene **a** produces defective hemoglobin which causes sickle-cell anemia.



Try This!

Use Punnett squares to determine the chances for each of the following sets of parents having a child with sickle-cell anemia.

- (a) one parent carries the gene ($Aa \times AA$)
- (b) both parents carry the gene ($Aa \times Aa$)
- (c) one parent carries the gene and the other has the disease ($Aa \times aa$)

For each set of parents, what is the chance of having a child with sickle-cell anemia?

Exposure to environmental factors can also cause mutations. Environmental factors that cause mutations are called **mutagens**. These include radiation; ultraviolet light; and many chemicals, including PCBs, those in cigarettes, and alcohol and other drugs.

Most of these errors do not pass from one generation to the next. There are exceptions. For example, children may pass on some effects from their mother's exposure to radiation.

Inherited Genetic Disorders

Sometimes, a mutation may be helpful. For example, giraffes with a mutation for a longer neck increase their ability to obtain food. Usually mutations result in **genetic disorders**, which result from alterations to the genes. For example, sickle cell anemia results from an abnormal form of hemoglobin. Hemoglobin is the substance in red blood cells that carries oxygen through the body. Someone with sickle cell anemia has abnormal hemoglobin. This leads to poor resistance to infection and low levels of iron in the blood.

Genetic Disorders Caused by Mutagens

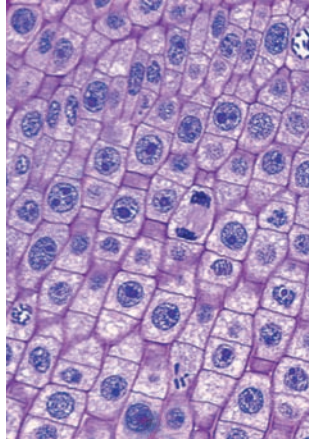
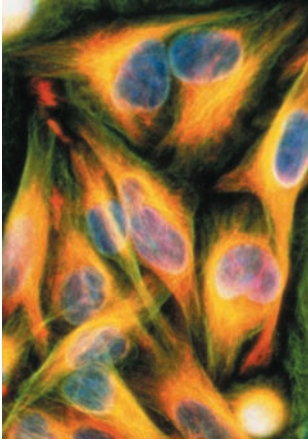


Figure 12.8 Radiation alters the DNA in normal cells to form cancer cells. The cancer cells keep dividing and build up masses of tissue called a tumour.

Radiation from X-ray machines and ultraviolet light in sunlight or tanning lamps can cause mutations. Such radiation releases particles smaller than atoms into cells and causes chromosomal changes. An overdose may cause cancer, skin burns, and other serious conditions.

Doctors use radiation beneficially to treat diseases such as cancer. They focus large doses of radiation on the cancer cells in an attempt to destroy the cell nuclei. Damage to non-cancerous tissue is limited by shielding the healthy tissue from radiation. The amount of radiation is also carefully monitored.

Human embryos are particularly at risk from exposure to mutagens. Table 12.2 examines some dangerous chemicals and their effects on human embryos.

Table 12.1 Mutagens and Effects on Human Embryos

Mutagen	Effect
Exposure to alcohol Children born to mothers who consume any alcohol during their pregnancy may suffer from a condition called fetal alcohol spectrum disorder (FASD).	- low birth weight - facial and joint abnormalities, small heads - heart defects - difficulties with learning, behaviour, and mental health
Exposure to prescription and illegal drugs Pregnant women should take prescription drugs only under a doctor's care. Doctors who know the woman is pregnant look for medications that do not harm the fetus.	- premature births - low birth weight - drug addiction at birth - possible mental limitations - possible birth defects
PCBs (polychlorinated biphenyls) PCBs are a mixture of toxic chemicals present in the environment because of improper disposal of wastes and consumer products. Some are linked to various forms of cancer, including liver and kidney cancer.	- low birth weight - mild symptoms that often disappear with age: anemia, liver conditions, skin rashes - difficulties co-ordinating movement and problems with short-term memory that last several years - delays in normal development - problems with immune system

READING check

Name three mutagens that harm human embryos and describe the effect of each.

Key Terms

mutation
mutagens
genetic disorders

Check Your Understanding

- Briefly describe how a mutation might occur when chromosomes are copied. Use a sketch if you wish.
- (a) What are mutagens?
(b) Name one example of a mutagen and identify a genetic disorder that results from exposure to it.
- Why is it important that pregnant women avoid consuming any amount of alcohol or taking any type of drug?

12.4 Issues in Genetic Research

Many people believe that the Human Genome Project is the most important scientific effort that humans have ever attempted. All of the genes found in a complete set of chromosomes — the set of genetic instructions that makes you what you are — form a genome. Scientists working on the genome project have mapped where each gene belongs on 46 human chromosomes.

This kind of information has many benefits. One benefit is that scientists can use this knowledge to transfer genes from one organism to another to give one organism the traits of another. Transferring genes from one organism or type of organism to another is referred to as **genetic engineering**. For example, scientists have inserted a spider gene into goats. The goat then produces a protein identical to the protein found in spider silk. The protein is removed from goat milk to produce silk fibres used to make bulletproof vests.

Someone who suspects they have a genetic disorder such as a genetic disposition toward cystic fibrosis or breast cancer, may use genetic research to check their chances of developing a specific disease. They may contact a genetic counsellor, who gathers medical information about the individual's family and analyzes the person's chromosomes to help predict the chances of the person developing a particular disorder.

Table 12.3 outlines some of the benefits of these advances in genetic research and engineering. But there are also concerns, which you will learn about on the next page.

DidYouKnow?

This pregnant woman is having an **amniocentesis**. This procedure is used to determine the chromosomal make-up of a fetus. The test shows whether a fetus may have a genetic abnormality. Some parents use such tests to decide whether or not to continue with a pregnancy. This picture shows ultrasound being used to locate the fetus. Doctors need to know where the fetus is so that they do not damage it when they insert a large hollow needle to withdraw amniotic fluid.



Figure 12.9

Table 12.2 Benefits of Genetic Research

Benefit	Examples
identifying the source of thousands of genetic disorders (such as colourblindness, hemophilia, or cystic fibrosis)	If scientists can identify the gene that causes a disease such as epilepsy, they may be able to alter that gene and thus cure the condition.
developing new drugs and treatments	Researchers identified the human DNA responsible for insulin production. They inserted that DNA into bacteria. Each time the bacteria cells divide, the human insulin gene is duplicated. The technology produces an inexpensive supply of insulin for people who have diabetes.
helping people make lifestyle choices	By studying a family's genes, researchers can determine which diseases people with that genetic make-up are most likely to develop. Someone with a family history of heart disease, for example, might take special care to eat a low fat diet and exercise regularly.



Identify at least three benefits of genetic research.



Figure 12.10 Scientists do not know all the risks of new technologies. For example, Dolly, the world's first cloned sheep, developed arthritis before she was six years old. Researchers do not know whether Dolly's arthritis was caused by environmental factors or genetic factors.

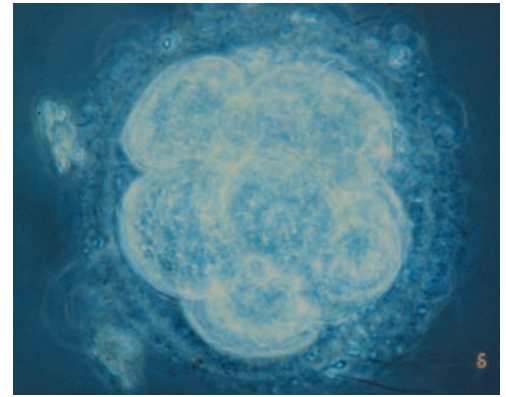


Figure 12.11 Many people question the ethics of using a human embryo such as the one shown here to harvest stem cells. Stem cells are unspecialized cells that are the building blocks for many body parts. Researchers hope to use stem cells to treat a variety of diseases, including Alzheimer's disease, and grow new organs. Is it right to kill an embryo in order to do this?

Concerns about Genetic Research

What would happen if all parents decided to have only male — or only female — children? This is the kind of question people ask when discussing genetic research. Such questions are ethical questions. They consider what is right or wrong.

Some ethical concerns deal with questions about who should have access to genetic information. For example, might employers ask for genetic information about employees? How might they use it?

Consider the ethical issues below, and then identify the ethical concerns raised in Figures 12.10 and 12.11.

- Who has a right to genetic data that provide information about your chances of developing various conditions? What might happen if health insurance companies had access to everyone's genetic make-up? Could they refuse to provide some people with health insurance? What might happen if an employer found out that an employee had genes that often result in substance abuse? Might the employee be fired?
- Do you want to know genetic information that may tell you of a terrible disease you are likely to develop but unable to do anything about? How might knowing this affect the way you live? Might you suffer from depression or adopt behaviours that put you more at risk?

READING
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Identify two concerns about advances in genetic research.

Check Your Understanding

1. How might amniocentesis create an ethical problem for some people?
2. Working with a partner, list three pros and cons of using advances in genetics. Develop a chart similar to the one shown here to help you organize your information.
3. Choose one ethical dilemma related to genetic research. Work with a partner to discuss the points of view on this dilemma. Consider how you feel about the issue. Express your point of view orally.

Pro	Con

Key Terms

genetic engineering
amniocentesis